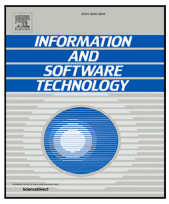
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# Metamorphic testing for textual and visual entailment: A unified framework for model evaluation and explanation

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ABSTRACT

**Context:** Textual entailment (TE) and visual entailment (VE) serve as the basis for a broad spectrum of tasks in natural language processing and vision–language modeling. However, although being extensively studied, both TE and VE models exhibit several quality issues. Additionally, their black-box nature hampers the understanding of their behaviors, making it unclear why the model fails to correctly predict entailment relationships. Consequently, there is a pressing need for methods that can effectively evaluate and explain TE and VE models.

**Objective:** This study aims to develop a unified approach for detecting and interpreting failures, in both TE and VE models.

**Methods:** We propose a metamorphic testing-based approach for evaluating and explaining both TE and VE models. The central aspect of our approach lies in the proposed three metamorphic relations, which are generic to both TE and VE, and also preserve specific associations among relevant inputs. The proposed approach conducts metamorphic testing to detect failures in TE and VE models. When a failure is revealed, it further performs a post-hoc analysis within the relevant group of inputs to identify information that is critical for the detected failure.

**Results:** Experimental results demonstrate the effectiveness of the proposed approach in failure detection and also confirm its potential to provide useful information to pinpoint the root causes of detected failures.

**Conclusion:** This study presents a general metamorphic testing approach for both TE and VE. It also demonstrates that, with specifically designed metamorphic relations, metamorphic testing can serve as an effective basis for model explanation.

## 1. Introduction

Reasoning is a fundamental capability in artificial intelligence (AI), enabling models to draw logical inferences and interpret information [1]. Textual entailment (TE) [2] and visual entailment (VE) [3] are representative reasoning tasks towards natural language processing (NLP) and vision–language modeling, respectively. Specifically, TE determines whether a given hypothesis logically follows from a textual premise. VE, in contrast, extends this reasoning to multimodal settings, where an image serves as the premise, and the hypothesis is a textual description. Both TE and VE have been a fundamental basis for a series of downstream applications [2,4]. TE is essential for tasks such as question answering, fact checking, and information retrieval. VE is also important for various vision–language tasks, enabling applications like visual question answering and cross-modal inference.

Due to their central role in advancing reasoning capabilities across NLP and vision–language models, the quality of TE and VE models is highly concerned. However, despite their advancements, both TE and VE models exhibit various quality issues, including spurious correlations, dataset biases, and brittle reasoning patterns [5,6]. More seriously, some models rely on superficial cues rather than true logical inference, leading to erroneous entailment predictions in real-world applications [7]. Therefore, evaluation methods are needed to assess the correctness and robustness of TE and VE models.

Existing evaluation methods rely on benchmark datasets and human-annotated test sets to provide performance measurement for TE and VE. However, these methods always require labeled ground truth for each sample, which can be costly and time-consuming to create, thus limiting their applicability and scalability. Metamorphic testing (MT) [8,9] is a property-based testing method, which is effective in alleviating the oracle problem. Due to the independence of ground truth labels, MT has been an important method for evaluating and assessing

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intelligent models, including VE and TE models. Nevertheless, although feasible and effective, existing MT methods focus on merely TE or VE, but not both of them [10,11]. This limitation highlights the need for a general MT-based evaluation method that can be adapted to both TE and VE.

On the other hand, the black-box nature of TE and VE models hinders the understanding of their behaviors. More specifically, when a failure occurs, it is often difficult to pinpoint the exact cause. This lack of interpretability makes it unclear why the model fails to correctly predict entailment relationships, which in turn impedes model debugging and enhancement. Therefore, a systematic method for explaining the root causes of failures in TE and VE models is highly desirable.

To address the above issues, this study proposes an MT-based approach for evaluating and explaining both TE and VE models. Our key insight is that metamorphic relations (MRs) focusing on the commonality of TE and VE can increase their adaptability, and that analysis within individual groups of test inputs revealing failures can help to pinpoint the exact cause. Our approach, MTEE, is built upon three hypothesis-centric MRs supported by text decomposition operation, including one MR directly adopting hypothesis decomposition, as well as two composite MRs derived from it. On one hand, the proposed MRs restrict the input operation merely to the textual hypothesis, which is a common part of the input of TE and VE. This guarantees the adaptability to both TE and VE. Additionally, the proposed MRs are equipped with hypothesis decomposition operation, breaking down an original hypothesis into a set of sub-hypotheses. Since the reasoning of an entire hypothesis is highly related to the reasoning of all of its sub-hypotheses, these MRs facilitate a post-hoc analysis that extracts critical information for explaining detected failures.

We conduct comprehensive experiments to evaluate MTEE, on two typical TE models (ESIM [12] and DeBERTa [13]) and two state-of-the-art VE models (OFA-VE [14] and ALBEF-VE [15]), using the SNLI [16] and SNLI-VE [3] datasets. Our results demonstrate that MTEE identifies a significant number of failures in the subject models, and that our proposed MRs outperform the baseline MRs in terms of failure detection effectiveness. Additionally, experimental findings confirm that MTEE can generate concise explanations, which are valuable for pinpointing critical information to interpret and understand detected failures. Our analysis also provides insights into the performance of the subject models and discusses key factors that influence their performance.

The key contributions of this study are summarized below.

- We propose an MT-based approach, MTEE, for evaluating and explaining TE and VE models.[1] Our approach is applicable to both TE and VE, and it facilitates failure detection as well as failure interpretation.
- Our MRs associate the reasoning of the original hypothesis with that of the relevant sub-hypotheses from the same premise. This association enables the identification of critical information in the original hypothesis to address the question: "Which words or phrases in the hypothesis lead to the revealed failure?"
- The experimental results show that MTEE has a promising capability for revealing failures, and is also effective in generating concise explanations to pinpoint critical information related to detected failures.
- Our results reveal distinct characteristics of subject models, and also clarify common factors affecting their performance.

The rest of this paper is structured as follows. Section 2 introduces background knowledge about TE, VE, and MT, and also clarifies the motivation of this study. Section 3 presents the technical details of our approach, including the proposed MRs, the failure detection method, and the failure explanation method. Section 4 clarifies our experimental setup, followed by a detailed experimental analysis in Section 5.

[1] https://github.com/BinTangHuLu/MTEE

| Premise | Hypothesis | Output |
|---|---|---|
| *Two young boys wearing shorts and sandals throw pebbles from a dirt path into a body of water.* | *Pebbles are being thrown into a body of water by two boys.* | Entailment |
| | *The body of water is a lake.* | Neutral |
| | *The two young boys are playing kickball at the park.* | Contradiction |

(a) Textual Entailment.

| Premise | Hypothesis | Output |
|---|---|---|
|  | *Some people at a wedding.* | Entailment |
| | *A large family poses for a photo for their Christmas card.* | Neutral |
| | *A small family is eating a meal.* | Contradiction |

(b) Visual Entailment.

**Fig. 1.** Examples for textual entailment and visual entailment.

Section 6 discusses the specificity and limitations of the proposed approach. Section 7 presents and discusses related work. Finally, Section 8 concludes this study and also discusses future research plans.

## 2. Background and motivation

### 2.1. Textual and visual entailment

The entailment task focuses on the inferential relationship between a pair of premise ($P$) and hypothesis ($H$). It aims to determine whether or not the information and facts conveyed by the hypothesis can be inferred from the given premise. The entailment task finally predicts a label for $P$ and $H$, which is either *entailment*, *contradiction*, or *neutral*.

(1) *Entailment*, denoted by $P \rightarrow H$, indicating that $H$ can be confirmed to be true when $P$ is true.
(2) *Contradiction*, denoted by $P \nrightarrow H$, indicating that $H$ is confirmed to be false when $P$ is true.
(3) *Neutral*, denoted by $P \perp H$, indicating that the truth of $H$ cannot be determined based on $P$.

Two typical categories of entailment tasks are textual entailment and visual entailment.

- Textual entailment (TE) [2], also known as natural language inference, is a fundamental research in the field of Natural Language Processing (NLP). It focuses on understanding the directional relationship between two texts, where the premise either supports, contradicts, or is neutral to the hypothesis (as illustrated in Fig. 1(a)). By bridging the gap between raw text and logical inference, TE plays a pivotal role in advancing NLP systems toward human-like language understanding.
- Visual entailment (VE) [3] is a multimodal inference task that involves both textual and image information (as illustrated in Fig. 1(b)). It extends the concept of TE by incorporating visual information, in order to determine whether a given image (the premise) logically supports, contradicts, or is neutral to a textual statement (the hypothesis). VE has been a basis for various applications, including visual question answering, image captioning, and multimodal reasoning, etc.

It is worth noting that both VE and TE focus on deciding the entailment relationship between the given premise and a textual hypothesis. However, TE focuses on a textual premise, while VE targets an image premise.

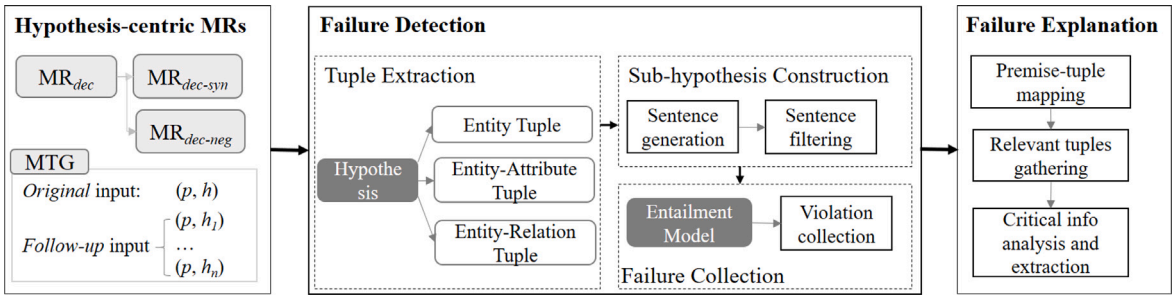



**Fig. 2.** Overview of MTEE.

### 2.2. Metamorphic testing

Metamorphic testing (MT) is a property-based testing method [8,9], with a broad range of applications. MT employs metamorphic relations (MRs) to describe some properties of the system under test (SUT), via relationships among relevant inputs and their outputs. Specifically, an MR specifies the way of transforming the *original* input to construct relevant *follow-up* inputs, and also describes the expected relationship between the original and follow-up outputs.

The original and its corresponding follow-up inputs of an MR constitute a metamorphic test group (MTG). Once original inputs are prepared and MRs are identified, a set of MTGs can be constructed according to each MR. Based on these, MT executes individual MTGs on SUT, and further validates the relationships among original and follow-up outputs against the relevant MR. A *violating* MTG, with the original and follow-up outputs failing to satisfy the relationship specified by the MR, reveals failures of SUT via an MR violation. One critical characteristic of MT is that it can be independent of the test oracle, since it judges test outcomes by using MRs, rather than relying on the test oracle. Because of this, MT can alleviate the test oracle problem, which is a fundamental challenge of software testing.

Apart from testing, MT has been further extended to software validation [17,18] and quality assessment [19], and has also been integrated into other software engineering activities, such as fault localization [20] and program repair [21]. In recent years, MT has been applied to various NLP and multimodal tasks, including visual question answering [22], image captioning [23], machine translation [24], as well as TE [10] and VE [11].

### 2.3. Explaining black-box models

Advanced models are inherently complex and opaque [25]. A model often operates through intricate, non-linear computations that are difficult to trace, leading to opaque decision-making processes (which is known as the black-box nature). This lack of transparency hinders the understanding of the rationales behind predictions or decisions, potentially undermining trust and introducing operational risks [26]. Consequently, there is increasing interest in developing explanation approaches that can make intelligent models more interpretable and comprehensible.

Generally, explanations are either *global* or *local*. Global explanations aim to provide a holistic understanding of a model's overall behavior, while local explanations focus on interpreting the model's behavior for individual predictions. Furthermore, explanation approaches can be broadly classified into intrinsic and post-hoc methods. Intrinsic explanations are derived from models that are inherently interpretable by design. Differently, post-hoc methods aim to approximate or reveal the reasoning behind a model's predictions without modifying its internal structure. This study focuses on locally explaining the failures of black-box entailment models.

### 2.4. Motivation

Despite significant advancements in both TE and VE, driven by deep learning architectures, these models are not infallible and may still produce incorrect entailment predictions, showing inference failures. Such failures can stem from various factors, including ambiguous language, complex visual scenes, or limitations in the model's ability to fully capture contextual and multimodal relationships. Although MT has been applied to entailment tasks, existing approaches typically focus on either TE or VE individually rather than addressing both simultaneously.

On the other hand, the black-box nature of both TE and VE models adds another layer of complexity. When failures occur, it is often challenging to understand their root causes due to the lack of interpretability. Understanding why a model made a wrong prediction is crucial for improving its performance and ensuring transparency. Existing MT-based evaluation methods, however, have not tackled the problem of explaining the root cause for failures of VE and TE.

The above observations motivate this study to focus on a unified approach for both TE and VE, facilitating not only the detection of failures, but also explaining why the failure is incurred.

## 3. Approach

This section presents our approach, MTEE, for evaluating and explaining TE and VE. We first introduce the key idea and overall workflow of MTEE (Section 3.1). Then, we present the rationale and details of three MRs that are generic to entailment tasks (Section 3.2), followed by the description of procedures respectively for inference failure detection (Section 3.3) and failure interpretation (Section 3.4).

### 3.1. Overview of MTEE

MTEE employs MT to detect inference failures of TE and VE models, and further interprets the root causes behind each failure via a post-hoc analysis. To ensure applicability across various entailment tasks, MTEE adopts hypothesis-centric MRs, which operate only on the hypothesis of the original input to construct follow-up inputs, such that an original input and its corresponding follow-up inputs share the same premise but incorporate different hypotheses. On the other hand, inspired by the divide-and-conquer idea and the pipeline of entailment task, the relationship between the original premise and hypothesis should be closely linked to the relationships between the premise and its corresponding sub-hypotheses. Based on this observation, MTEE equips hypothesis-centric MRs with input operations driven by text decomposition.

Fig. 2 shows the overall workflow of MTEE. With the guidance of hypothesis-centric MRs, MTEE first constructs MTGs by making use of existing datasets. Then, it conducts MT to collect inference failures of entailment models. After that, MTEE further performs a post-hoc analysis with respect to relevant MRs, in order to identify critical information in the hypothesis that is likely to cause inference failures.

### 3.2. Generic metamorphic relations for TE and VE

One of MTEE's objectives is to ensure adaptability across a broad range of entailment tasks. With regard to the entailment tasks, although the premise may vary in type (e.g., text or image), they all focus on a textual hypothesis. Moreover, because a hypothesis can be decomposed into a series of sub-hypothesis, such that the relationship between the original premise and hypothesis is associated with the relationship between the same premise and the corresponding sub-hypotheses. Accordingly, we identify hypothesis-centric MRs that manipulate merely the hypothesis of inputs based on hypothesis decomposition.

We use $t_s = (p, h)$ to denote the original input, where $p$ and $h$ are, respectively, the premise and hypothesis. We use $t_f = (p, h')$ to represent the corresponding follow-up input. It is noted that with the use of hypothesis-centric MRs, premise-hypothesis pairs of $t_s$ and $t_f$ differ only in the hypothesis part (that is, $h$ and $h'$). Therefore, we refer to $h$ and $h'$ as the original and follow-up hypothesis, respectively. We also use $E(p, h)$ to denote the label predicted by the target entailment model (say $E$) for the given input $(p, h)$.

We first introduce one MR that constructs the follow-up input by merely decomposing the original hypothesis. Our key insights are twofold. Firstly, as a textual sentence, the hypothesis typically contains multiple details and diverse types of information, for which the decomposition is feasible. Secondly, the entailment model assesses whether all information in the hypothesis can be inferred from the given premise. This means that each individual part of the hypothesis must be inferred, collectively contributing to the overall entailment outcome.

Accordingly, we use *Decompose* to represent the operation that breaks down a hypothesis into a set of sub-hypotheses. That is, $\{h_1, h_2, \ldots, h_n\} = Decompose(h)$. The MR is described as follows.

**MR**$_{dec}$: Suppose that $t_s = (p, h)$, where $p \rightarrow h$. $h' = Decompose(h)$, yielding $t_f = (p, h') = \{ (p, h_1), (p, h_2), \ldots, (p, h_n)\}$. Then, $\forall h_i$, $E(p, h_i) = E(p, h)$.

To make MR$_{dec}$ a good basis for testing and explaining entailment models, *Decompose* is expected to derive sub-hypotheses having the following properties.

- *Compatibility.* Each sub-hypothesis should be compatible with the original hypothesis, ensuring that the information it conveys aligns seamlessly with the original hypothesis. The transitive property of logical inference permits deducing $p \rightarrow h_i$, given $p \rightarrow h$ and $h \rightarrow h_i$, where the compatibility of sub-hypotheses is critical for establishing $h \rightarrow h_i$. This property is essential to maintain the validity of MR$_{dec}$.
- *Completeness.* All sub-hypotheses collectively represent the entire semantics of the original hypothesis. This property enables the effectiveness of MR$_{dec}$ in detecting failures of entailment models.
- *Atomicity.* Each sub-hypothesis should be concise and contains a small amount of information. This property facilitates information isolation and will make MR$_{dec}$ helpful for interpreting failures of entailment models.

We further identify two MRs by following the MR composition strategy [27]. Our previous study [10] has proposed several MRs for TE. With further analysis, we select two MRs, MR$_{syn}$ and MR$_{neg}$ (as detailed in Section 4.4), that are able to support hypothesis-centric operation, and can also be combined with MR$_{dec}$. We use *Synonym* to denote the operation of replacing a word in a given text with its synonym, and use *Negation* to represent the operation of negating a given sentence. The two composite MRs are described as follows.

**MR**$_{dec\text{-}syn}$: Suppose that $t_s = (p, h)$, where $p \rightarrow h$. $h' = Synonym(Decompose(h))$, yielding $t_f = (p, h') = \{(p, h'_1), (p, h'_2), \ldots (p, h'_n)\}$. Then, $\forall h_i$, $E(p, h'_i) = E(p, h)$.

**MR**$_{dec\text{-}neg}$: Suppose that $t_s = (p, h)$, where $p \rightarrow h$. $h' = Negation(Decompose(h))$, yielding $t_f = (p, h') = \{(p, \neg h_1), (p, \neg h_2), \ldots, (p, \neg h_n)\}$. Then, $\forall h_i$, $E(p, \neg h_i) = \neg E(p, h)$.

Specifically, $Synonym(Decompose(h))$ means applying the synonym replacement operation on every sentence resulting from the decomposition. That is, $h'_i = Synonym(h_i)$. Similarly, $Negation(Decompose(h))$ indicates negating each sentence $h_i$ to generate $\neg h_i$.

### 3.3. Inference failure detection

MTEE conducts MT to reveal inference failures of entailment models. Guided by hypothesis-centric MRs, it first breaks down the original hypothesis into a set of sub-hypotheses. Then, it constructs MTGs according to individual MRs, and finally executes MTGs and checks their outputs against the relevant MRs (as outlined in Fig. 3).

#### 3.3.1. Hypothesis decomposition

To facilitate failure detection and explanation, MTEE aims to decompose a given hypothesis into a *complete* set of *atomic* sub-hypotheses that are *compatible* with the original hypothesis. To do this, the hypothesis decomposition consists of the following three steps.

*Step 1. Tuple extraction.* To ensure atomicity and also increase completeness, MTEE first extracts structured tuples that carry simple facts and semantics, from the given hypothesis. Regarding the characteristics of entailment tasks, we design three types of tuples to capture essential semantic information: (1) ⟨E⟩, the entity tuple, which consists of a noun or noun phrase representing a concrete or abstract entity. (2) ⟨E, A⟩, the entity-attribute tuple, which describes an entity with specific attributes or properties. (3) ⟨E, R, E'⟩, the entity-relation tuple, which expresses a specific type of relationship between entities.

*Step 2. Sentence generation.* For each tuple obtained in *Step 1*, MTEE further generates a simple sentence from it. Each generated sentence is expected to be a sub-hypothesis that is consistent with the original hypothesis.

*Step 3. Sentence filtering.* To guarantee compatibility, MTEE filters sub-hypotheses that take inconsistent facts or information. This is achieved by validating the entailment relationship between the original hypothesis and each of its sub-hypotheses. That is, for any sub-hypothesis $h_i$, it is regarded to be compatible with the original hypothesis $h$ only if $h \rightarrow h_i$. To do this, for each sub-hypothesis generated in *Step 2*, MTEE employs an entailment model (the model under test is utilized for TE tasks, while the DeBERTa model is applied for VE tasks) to predict whether or not the original hypothesis entails it. If yes, it is a compatible sub-hypothesis and will be kept for failure detection and explanation. Otherwise, such a sub-hypothesis will be filtered out. Consider the original input of MR$_{dec}$ depicted in Fig. 3 as an example. For the given hypothesis $h$, five sub-hypotheses (h1 to h5) are constructed, while one of them, h3, contradicts with $h$ (as evidenced by $h \not\rightarrow$ h3). Noted that h3 is resulted from improper and insufficient processing of $h$. Consequently, h3 is filtered out and the final follow-up inputs are ($p$, h1), ($p$, h2), ($p$, h4), and ($p$, h5). The exclusion of ($p$, h3) is necessary, because such a follow-up input breaks the fundamental constraints of the MR ($p \rightarrow$ h3 is non-deducible when $p \rightarrow h$ but $h \not\rightarrow$ h3), which will introduce noise into the testing procedure.

Hypothesis decomposition requires advanced natural language processing for both tuple extraction and sentence generation to produce accurate and linguistically natural sub-hypotheses. However, this is particularly challenging, since original hypotheses may vary in complexity, style, or contain syntactic/semantic irregularities [16]. Moreover, MTEE is designed for black-box models, it is thus expected to be generalized to a broad range of entailment models, especially for those handling different types of languages. Given these requirements and also inspired by the impressive performance of large language models, in this study, we leverage ChatGPT for both tuple extraction and sentence generation.

We design structured prompt templates for both tasks, as shown in Fig. 4. Each template mainly consists of four parts. (1) *Task definition*, which clearly outlines the requirement for tuple extraction or sentence

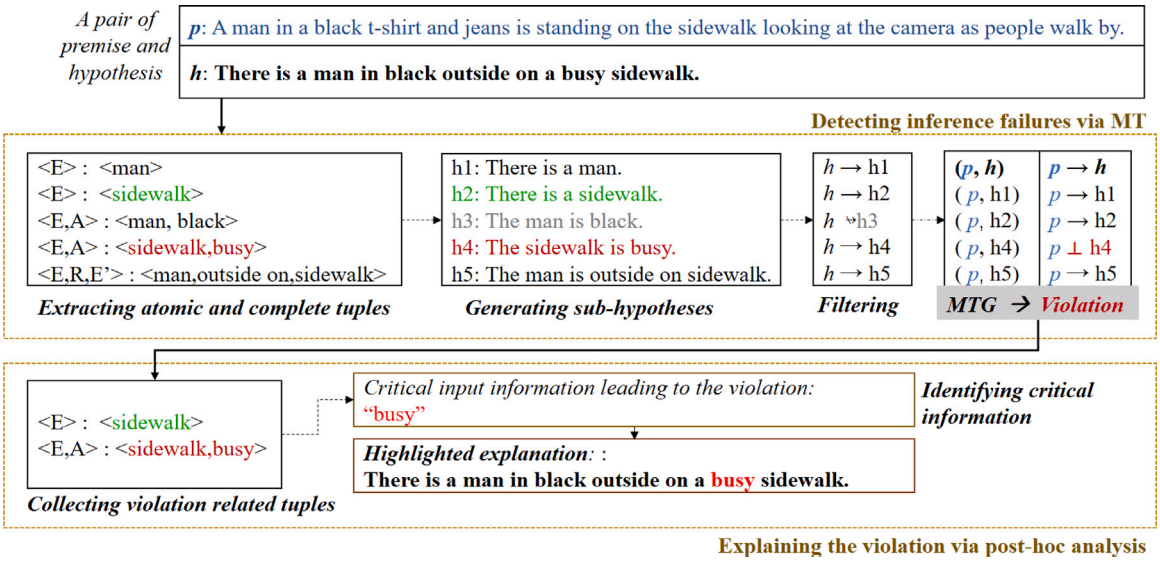



**Fig. 3.** The workflow of MTEE to detect and explain inference failures. The incompatible sentence to be filtered out is marked with gray color, the sub-hypothesis that leads to a violation as well as the corresponding tuple, are marked with red color, while the relevant sub-hypothesis and tuples satisfying the MR, are marked with green color.

generation. (2) *Output constraints*, which specify the formats or properties of the output. (3) *Input*, which is the raw text to be processed. (4) *Few-shot examples*, which provide several input–output examples to guide ChatGPT in task execution. Both tasks are accomplished by ChatGPT 3.5, with a temperature of 0.1, which has been found to be a reasonable temperature for various NLP tasks [28,29].

#### 3.3.2. Inference failure collection based on MRs

Upon the construction of sub-hypotheses, MTEE constructs MTGs and conducts MT, according to the three MRs. For an MTG of $MR_{dec}$ and $MR_{dec\text{-}syn}$, if at least one follow-up output is not identical to the original output, an MR violation is detected, and this violating MTG reveals a failure of the model under test. In contrast, for an MTG of $MR_{dec\text{-}neg}$, if there exist some follow-up outputs that are identical to the original output, the MR is violated on this MTG, indicating a failure of the model under test.

In this study, Negate [30],[2] a lightweight and open-source tool for sentence negation, is employed to support the construction of MTGs of $MR_{dec\text{-}neg}$. Moreover, SpaCy,[3] an open-source natural language processing library, is used to support $MR_{dec\text{-}syn}$.

### 3.4. Inference failure explanation

Once an MR violation is detected, MTEE conducts a post-hoc analysis to identify the root cause of the violation in relation to the corresponding MTG. Specifically, it determines a subset of elements (i.e., words or phrases) in the original hypothesis that contribute to the inference failure, leading to the MR violation. This process effectively answers the question: "Which aspects of the original hypothesis influence the model's decision, ultimately resulting in an MR violation?"

Our three MRs are based on hypothesis decomposition, generating MTGs that consist of the original hypothesis and a set of atomic sub-hypotheses (i.e., follow-up hypotheses). In a violating MTG, only some of the sub-hypotheses exhibit deviations in entailment results, leading to an MR violation. The analysis of the sub-hypotheses — those that exhibit violations and those that satisfy the MR — enables the identification of the critical information in the original hypothesis that contributes to the MR violation. As shown in Fig. 3, among the sub-hypotheses h1, h2, h4, and h5, only h4 violates the MR. By analyzing the tuple of h4 alongside the tuple of h2, which is closely related to h4, we can pinpoint the critical information that causes the MR violation.

The detailed procedure is described by Algorithm 1, and also illustrated in Fig. 3. Given a violating MTG and an MR, the algorithm first classifies all sub-hypotheses into two groups, those violating the MR and those satisfying the MR (line 2 to line 3). Then, the relevant tuples of these two groups of sub-hypotheses are collected (line 4–line 5). After that, each violation-related tuple, say $u_i$, is analyzed. The critical information is identified according to the following principles (line 9). (1) If $u_i$ belongs to ⟨E⟩, then the entity information will be identified; (2) If $u_i$ belongs to ⟨E, A⟩, the entity information will be identified if the relevant ⟨E⟩ does not leading to MR violation. Otherwise, both entity and attribute information will be identified. (3) In the case that $u_i$ belongs to ⟨E, R, E'⟩, the entire tuple information will be identified if the relevant ⟨E⟩ and ⟨E'⟩ lead to MR violations. Otherwise, the information of R will be identified if neither ⟨E⟩ and ⟨E'⟩ incur an MR violation, while the information of both E (E') and R will be identified if only E' (E) leads to no violation. At last, all identified information related to the given violating MTG will be combined and returned.

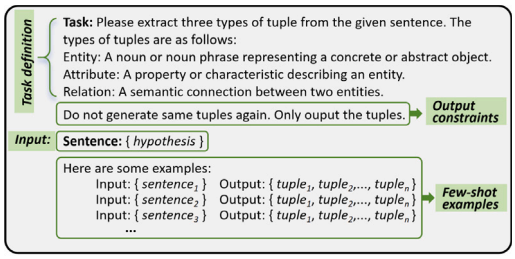
## 4. Experimental setup

### 4.1. Research questions

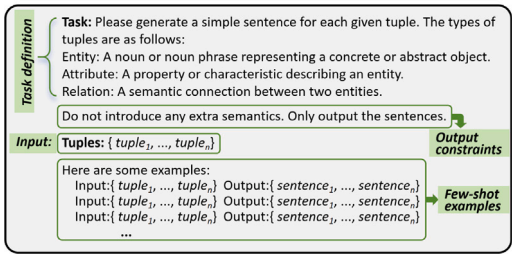
We evaluate the effectiveness of MTEE in detecting inference failures and explaining the root causes of failures by conducting extensive experiments. Our experiments focus on the following research questions.

- **RQ1**: *How does MTEE perform in detecting inference failures of TE and VE models?* This RQ investigates the effectiveness of MTEE in evaluating TE and VE models. We quantitatively report the violations detected by MTEE, and further compare our proposed MRs with baseline MRs. We also investigate the complementarity of the proposed MRs.
- **RQ2**: *Can MTEE effectively interpret the root causes of detected failures?* This RQ aims to study the capability of MTEE to generate explanations for clarifying the root cause of inference failures. We report the types of explanations, and also measure the compactness of them.

[2] https://github.com/dmlls/negate
[3] https://spacy.io/




(a) Prompt template for tuple extraction.




(b) Prompt template for sentence generation.

**Fig. 4.** Templates of ChatGPT prompt.

**Algorithm 1** Failure Interpretation

**Input:** $t_s = (p, h)$, $t_f = \{(p, h_1),\ldots,(p, h_n)\}$ // a violating MTG
$r_s$, $r_f\{r_1,\ldots, r_n\}$ // source and follow-up outputs
$R$ // the relevant MR
**Output:** $C$ // critical words for explanation

```
C = ∅
H_v = GetVioSubHypothesis(t_s, t_f, r_s, r_f, R)
H_s = GetSatSubHypothesis(t_s, t_f, r_s, r_f, R)
U_v = RetriveTuple(H_v);
U_s = RetriveTuple(H_s);
for each u_i in U_v do
    U_rv = ExtractRelevant(u_i, U_v)
    U_rs = ExtractRelevant(u_i, U_s)
    type,content = Analyze(u_i, U_rv, U_rs)
    C = C ∪ {type, content}
end for
return C
```

- **RQ3**: *What are the performance of subject entailment models?* In this RQ, we compare and investigate the performance of subject models, and also discuss important factors affecting their performance.

### 4.2. Subject models

We conduct experiments on four entailment models, including two TE models (ESIM [12] and DeBERTa [13]), and two VE models (OFA-VE [14] and ALBEF-VE [15]).

ESIM (Enhanced sequential inference model) [12] is a neural network architecture designed for TE. It leverages bidirectional LSTMs to encode input sentences, and applies attention mechanisms to capture word-level interactions. Known for its strong performance on typical datasets, ESIM has become a benchmark model in textual entailment. In this study, we utilize the PyTorch version of the pre-trained ESIM model from GitHub.[4]

DeBERTa (Decoding-enhanced BERT with Disentangled Attention) [13] is an advanced transformer-based model designed for TE. It improves upon BERT and RoBERTa by introducing disentangled attention and enhanced mask decoding architecture. DeBERTa has established itself as one of the strongest models for TE, often outperforming previous transformer-based architectures. In the experiments, we download the pre-trained model from Hugging Face.[5]

OFA-VE (One-For-All Visual Entailment) [14] is a unified multimodal model leveraging the OFA framework to integrate vision and language understanding seamlessly. It is pre-trained on a diverse set of tasks, making it effectively capture fine-grained interactions between textual and visual inputs. OFA-VE has demonstrated state-of-the-art performance, making it a leading model for VE tasks. We obtain the pre-trained OFA model from ModelScope.[6]

ALBEF-VE (Align Before Fuse for Visual Entailment) [15] is a state-of-the-art multimodal model designed for VE, and it mainly focuses on aligning visual and textual representations before fusing them for joint reasoning. ALBEF-VE improves entailment reasoning by refining vision–language representations, positioning itself as a strong performer in the VE domain. Since ALBEF does not provide a pre-trained model for the VE task, we downloaded the model and fine-tuned it following the official instructions.[7]

### 4.3. Test data

We employ two benchmark datasets, SNLI [16] and SNLI-VE [3]. SNLI is a widely used dataset for TE task, which contains textual data samples reflecting everyday situations and commonsense logic. Each data sample contains a premise-hypothesis sentence pair and their corresponding inference relationship. SNLI contains a training set with about 550,000 samples, and approximately 10,000 samples each for the validation and test sets. SNLI-VE is an extension of the original SNLI dataset, where each sample consists of an image premise and a natural language hypothesis. By including images from Flickr30k dataset, samples of SNLI-VE can reflect the interpreting of everyday scenes and activities involving people in diverse contexts (such as sports, work, leisure, and family life). The dataset contains 529,527 training samples, 17,858 validation samples and 17,901 test samples.

By using premise-hypothesis pairs from test sets of the two datasets, we further apply the proposed MRs to construct MTGs for supporting MTEE. As described in Section 3.2, the three MRs concentrate on source inputs where the premise entails the hypothesis. For a given premise-hypothesis pair, different models may predict varying relationships. As a result, the MTG sets for different subject models are different. Based on SNIL dataset, 10,455 MTGs are used for ESIM, while 10,067 MTGs are constructed for DeBERTa. Based on SNLI-VE dataset, 12,036 and 18,546 MTGs are constructed for OFA-VE and ALBEF-VE, respectively.

### 4.4. Baseline MRs

Some MRs have been previously proposed for evaluating TE or VE. Change et al. [11] proposed MRs for VE, while these MRs involve the transformation of both image premise and textual hypothesis, and thus cannot be adapted to TE. Our previous study [10] has proposed some MRs for TE, some of which can be adapted to support hypothesis-centric operations and thus be applicable to both TE and VE. With a detailed analysis, we select two MRs, denoted by $MR_{syn}$ and $MR_{neg}$, which are MR-$SS_3$ and MR-NR in [10]. These two MRs are employed as a comparison baseline. The following is a brief description of these two MRs.

[4] https://github.com/coetaur0/ESIM

[5] https://huggingface.co/cross-encoder/nli-deberta-v3-base

[6] https://www.modelscope.cn/models/iic/ofa_visual-entailment_snli-ve_large_en/

[7] https://github.com/salesforce/ALBEF

**Table 1**
Violation rates (violated MTGs/all MTGs) of our MRs and baseline MRs.

| Entailment models | | Baseline MRs | | MRs of MTEE | | | |
|---|---|---|---|---|---|---|---|
| | | $MR_{neg}$ | $MR_{syn}$ | $MR_{dec}$ | $MR_{dec\text{-}neg}$ | $MR_{dec\text{-}syn}$ | All |
| TE | ESIM | 6.96% (234/3358) | 10.01% (963/9612) | 13.52% (477/3527) | 14.17% (500/3527) | 48.57% (1652/3401) | 25.14% (2629/10455) |
| | DeBERTa | 6.33% (205/3235) | 8.58% (825/9612) | 10.36% (352/3395) | 10.98% (372/3395) | 38.02% (1246/3277) | 19.56% (1970/10067) |
| VE | OFA-VE | 50.22% (1919/3821) | 11.71% (2014/17195) | 11.89% (485/4079) | 65.99% (2692/4079) | 40.07% (1554/3878) | 39.30% (4731/12036) |
| | ALBEF-VE | 13.14% (782/5947) | 10.44% (1796/17195) | 25.92% (1621/6253) | 36.54% (2285/6253) | 38.07% (2300/6040) | 33.46% (6206/18546) |

**Table 2**
95% confidence intervals of violation rate differences between our proposed MRs and baseline MRs across TE and VE models.

| Entailment models | | $MR_{dec}$-$MR_{neg}$ | $MR_{dec}$-$MR_{syn}$ | $MR_{dec\text{-}neg}$-$MR_{neg}$ | $MR_{dec\text{-}neg}$-$MR_{syn}$ | $MR_{dec\text{-}syn}$-$MR_{neg}$ | $MR_{dec\text{-}syn}$-$MR_{syn}$ |
|---|---|---|---|---|---|---|---|
| TE | ESIM | 5.15%, 7.98% | 2.22%, 4.78% | 5.76%, 8.68% | 2.86%, 5.47% | 39.75%, 43.45% | 36.79%, 40.33% |
| | DeBERTa | 2.73%, 5.37% | 0.65%, 2.99% | 3.24%, 5.98% | 1.20%, 3.56% | 29.80%, 33.55% | 27.68%, 31.23% |
| VE | OFA-VE | −40.17%, −36.48% | −0.89%, 1.25% | 13.61%, 17.91% | 52.73%, 55.79% | −12.34%, −8.02% | 26.75%, 29.98% |
| | ALBEF-VE | 11.40%, 14.15% | 14.26%, 16.65% | 21.92%, 24.87% | 24.83%, 27.39% | 23.46%, 26.42% | 26.32%, 28.98% |

**$MR_{syn}$**: Suppose that $w$ is a word that is exclusively contained in $h$ of the original input, and $w'$ is a synonym of $w$. $h'$ is a new hypothesis constructed from $h$ by replacing $w$ with $w'$. Then, the output of $(p, h)$ and $(p, h')$ are expected to be identical.

**$MR_{neg}$**: Suppose that $t_s = (p, h)$, where $p \rightarrow h$. $h$ is negated to generate $\neg h$. Then, the output of $(p, h)$ and $(p, \neg h)$ are expected to be opposite (that is, $p \not\rightarrow \neg h$).

To support a fair comparison analysis, in our experiments, we re-implement $MR_{syn}$ and $MR_{neg}$, by adopting the same sentence negation and synonym replacement operation as is used for $MR_{dec\text{-}syn}$ and $MR_{dec\text{-}neg}$ (as clarified in Section 3.3.2).

## 5. Results and analysis

### 5.1. RQ1 (The effectiveness of MTEE in detecting inference errors)

To evaluate the effectiveness of failure detection, we apply MTEE to the four subject models. For each model, we collect its *violation rate* for individual MRs, which is the percentage of violating MTGs that successfully reveal failures. A higher violation rate indicates that more failures have been detected.

Table 1 reports the detailed test result, including the exact number of detected failures and the violation rate of each subject model, under different MRs. In total, MT reports 15,536 violations, revealing a large number of failures of our subject models. Moreover, each MR successfully detects violations for all of the four subject models, with a violation rate ranging from 10.36% to 65.99%. These results suggest that MTEE is able to detect failures in TE and VE models.

#### 5.1.1. Effectiveness of MRs

Each of our proposed MRs exhibits varying failure detection capabilities for different subject models. For instance, the violation rate of $MR_{dec\text{-}neg}$ is 10.98% on DeBERTa, while it is 65.99% on OFA-VE. This suggests that our MRs are capable of assessing different models, demonstrating their degree of conformance to the relevant MR. On the other hand, different MRs always exhibit varying violation rates for the same subject model. As can be observed from Table 1, for ESIM, the three violation rates of our three MRs are 13.52%, 14.17%, and 48.57%. This further demonstrates that each of our MRs possesses a unique ability to detect failures of subject models.

We also compare our proposed MRs with the two baseline MRs. As shown in Table 1, our three MRs generally have a better failure detection capability than baseline MRs. More specifically, $MR_{dec}$ achieves higher violation rates than $MR_{neg}$ on three out of four models, while it delivers higher violation rates on all four models, as compared with $MR_{syn}$. For the other two MRs, $MR_{dec\text{-}neg}$ and $MR_{dec\text{-}syn}$, they significantly outperform $MR_{syn}$ ($MR_{neg}$) in terms of the failure detection capability, on all subject models. In particular, $MR_{syn}$ has the largest number of MTGs among all MRs, since it has no restriction on the original input. However, its violate rates are still lower than the proposed MRs. To statistically compare our MRs with baseline MRs, we employ bootstrap resampling (10,000 iterations) to compute 95% confidence intervals (CIs) for the difference in violation rates. A difference is considered statistically significant if its 95% CI excludes zero; otherwise, it is deemed non-significant. As shown in Table 2, our analysis reveals statistically significant differences in 23 out of 24 comparisons, demonstrating the substantial impact of our proposed MRs. This result suggests that our proposed MRs are effective for evaluating TE and VE models.

As stated in Section 3.2, two of our MRs, $MR_{dec\text{-}neg}$ and $MR_{dec\text{-}syn}$, are composite MRs. $MR_{dec\text{-}neg}$ is derived from $MR_{dec}$ and $MR_{neg}$, while $MR_{dec\text{-}syn}$ is derived from $MR_{dec}$ and $MR_{syn}$. Therefore, we further compare individual composite MRs with their corresponding component MRs. According to the test results reported in Table 1, $MR_{dec\text{-}neg}$ and $MR_{dec\text{-}syn}$ have better failure detection capabilities than $MR_{dec}$. Furthermore, $MR_{dec\text{-}neg}$ has much higher violation rates than $MR_{neg}$, so does $MR_{dec\text{-}syn}$, as compared with $MR_{syn}$. This result confirms that the composite MRs proposed in this study are more effective than their corresponding component MRs. These results align with previous observations regarding the effectiveness of composite MRs, as both theoretically and empirically demonstrated in [27]. Compared to the component MR, which uses only a single type of input operation, our proposed composite MRs apply dual operations to the original hypothesis. As a result, the follow-up hypotheses generated by the latter differ significantly more from the original hypothesis, than those produced by the former. Consequently, the inference process for the source and follow-up inputs of composite MRs is more likely to trigger divergent model behaviors, enhancing their potentials to uncover inference failures.

In addition, we investigate the precision of MTEE in detecting entailment failures. To this end, we manually inspect violations reported by MTEE to determine how many of them represent real failures [31]. Given the large number of violations (15,536 violations in total), we followed existing studies [32,33] to randomly sample some violations. Specifically, we randomly selected 100 violations per MR for each of the subject models. In total, 1200 violations are inspected independently by two of our authors, with inter-rater reliability confirmed by a Cohen's Kappa coefficient of 0.826, indicating a strong agreement. The results are promising: the three MRs achieve precision scores ranging from 95% to 97%, with the average precisions of $MR_{dec}$, $MR_{dec-neg}$, and $MR_{dec-syn}$ be 97%, 96.5% and 95.5%, respectively. This indicates

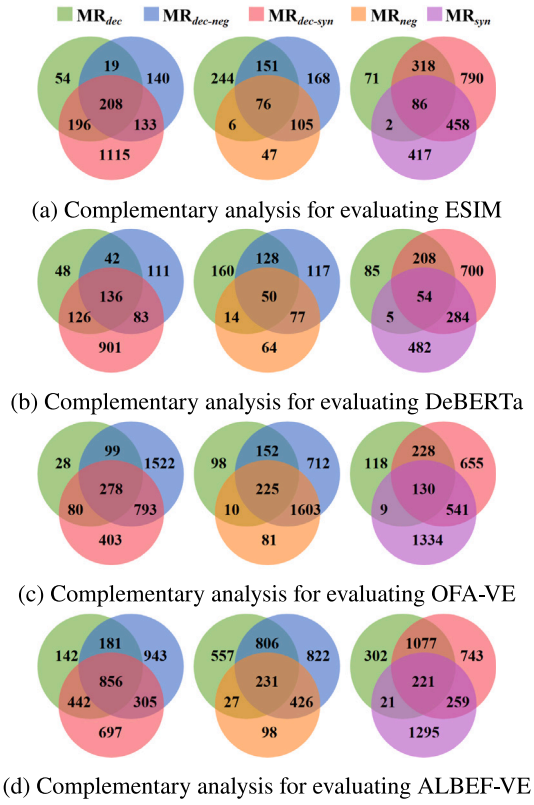



**Fig. 5.** Complementarity of MRs: Although different MRs exhibit varying violation rates, they are indispensable, as each of them exclusively reveals some errors.

that only around 5% of the reported violations are false positives, confirming MTEE's effectiveness in accurately identifying entailment failures.

#### 5.1.2. Complementarity of MRs

We further examine the necessity of each MR for evaluating the subject models. We seek to confirm whether an MR is indispensable because it complements other MRs, or unnecessary as its detected failures are entirely covered by other MRs.

When evaluating a given model, different MRs select original inputs from the same data set. Thus, if two violating MTGs from different MRs share the same original input, the failure identified is considered a common failure revealed by both MRs. Conversely, if a violating MTG includes an original input that does not appear in any other violating MTGs, the failure associated with this input can only be exclusively identified by the corresponding MR. We further investigate the number of common failures revealed by all MRs, the number of common failures revealed by individual pairs of MRs, as well as the number of distinct failures exclusively detected by each MR. To this end, we collect all violating MTGs for each of the MRs, with respect to different subject models. We conduct comparison among violating MTGs of three groups of MRs, (1) the proposed three MRs, (2) $MR_{dec\text{-}neg}$ and its component MRs, $MR_{dec}$ and $MR_{neg}$, and (3) $MR_{dec\text{-}syn}$ and its component MRs, $MR_{dec}$ and $MR_{syn}$.

Fig. 5 reports the results. It can be found that every of the MRs is indispensable for evaluating each of the subject models, because each MR can exclusively detect a certain number of errors. With a further investigation within each group, we have the following findings.

- Our proposed MRs are complementary to each other, while the two composite MRs detect significantly more exclusive errors compared to their component MR, $MR_{dec}$.
- $MR_{dec\text{-}neg}$ and its component MRs, $MR_{dec}$ and $MR_{neg}$, complement each other. Yet, $MR_{dec\text{-}neg}$ captures the highest number of exclusive errors, with $MR_{dec}$ ranking second, whereas $MR_{neg}$ detects only a small proportion of exclusive errors.
- $MR_{dec\text{-}syn}$ is mutually complementary to its component MRs, $MR_{dec}$ and $MR_{syn}$. $MR_{dec\text{-}syn}$ identifies a larger number of exclusive errors than $MR_{syn}$, for evaluating two TE models, while $MR_{syn}$ exclusively captures more errors for evaluating the VE models. The main reason for the superiority of $MR_{syn}$ is that it has a very large number of MTGs (e.g., 17195 vs 3878 for evaluating OFA-VE), most of which are ineligible to $MR_{dec\text{-}syn}$.

These results demonstrate that every of our MRs contributes to the evaluation of TE and VE models. Furthermore, our MRs also show superiority over the baseline MRs, in terms of being able to exclusively capture a large number of failures.

**Answer to RQ1.** MTEE has detected many inference failures in subject models. Compared to the baseline MRs, the proposed three MRs are of higher effectiveness. Furthermore, the proposed MRs are complementary with each other, and thus collectively contribute to the effectiveness of MTEE in evaluating TE and VE models.

### 5.2. RQ2 (The effectiveness of MTEE in explaining inference failures)

We further investigate the capability of MTEE to identify critical input information that is relevant to MR violations. MTEE can identify six categories of explanations, including Entity description (E), Attribute description (A), Entity accompanied with attribute description (EA), Relation description (R), Relation description involving one entity (ER), and Complete relation and entities description (ERE). Table 3 provides examples illustrating these types of explanations. Each explanation suggests that the relevant MR violation arises from an incorrect inference of the corresponding information in the hypothesis based on the premise.

For all of the 15,536 violations reported by MTEE, it further successfully generates explanations for 12,886 (82.9%) of them. Yet, it is also observed that MTEE generates multiple explanations for some of the violations. Table 4 summarizes the number of different types of explanations generated by using different MRs. It can be found that all three MRs are able to support the generation of the six types of explanations. These results demonstrate that MTEE can be effective in explaining the root causes of MR violations.

One key concern about an explanation refers to its compactness, that is, a good explanation is expected to contain a small amount of necessary information [26]. Since explanations for entailment tasks are textual, we further investigate the length of explanations via the number of words. Table 5 summarizes the average, minimal, and maximal length of explanations for four subject models, and Fig. 6 further shows the distribution of lengths of explanations. It can be observed that the explanations are notably shorter than the original hypotheses. On average, explanations provided by MTEE contain two or three words, and the majority of explanations have less than four words. These results suggest that MTEE performs well in generating compact explanations for TE and VE.

Noticing that MTEE fails to generate explanations for some violations, we manually inspect related data. There are mainly two cases. Firstly, for some violating MTG, the information extracted by MTEE covers the whole hypothesis. Such an explanation is ignored, since it is unable to pinpoint the root cause. On the other hand, since MTEE

**Table 3**
Examples of explanations generated by MTEE. $h$ represents the original hypothesis, while $h_i$, $h'_i$, and $\neg h_i$ respectively denote the follow-up hypothesis of $MR_{dec}$, $MR_{dec\text{-}syn}$, and $MR_{dec\text{-}neg}$.

| Type | MTG and entailment results | | | Explanation in highlight |
|---|---|---|---|---|
| | Premise ($p$) | Hypothesis: $h$, $h_i$, $h'_i$ or $\neg h_i$ | | |
| E | A man wearing a red shirt with the number 54 hits a baseball, while a catcher prepares to catch the ball. | $h$: A red shirted batter hits the pitch. | $(p \rightarrow h)$ | Words suggested: batter<br>(Original hypothesis with highlight explanation) A red shirted **batter** hits the pitch. |
| | | $h_1$: There is a batter. | $(p \perp h_1)$ | |
| | | $h_2$: There is a pitch. | $(p \rightarrow h_2)$ | |
| | | $h_3$: The batter is red shirted. | $(p \rightarrow h_3)$ | |
| | | $h_4$: The batter hits the pitch. | $(p \rightarrow h_4)$ | |
| A | A young female gymnast, wearing a sequined royal blue and white outfit, is lifting her right leg straight up in the air. | $h$: A younger gymnast, wearing dark blue and white, has her right leg straight in the air. | $(p \rightarrow h)$ | Words suggested: wearing dark blue and white<br>(Original hypothesis with highlight explanation) A younger gymnast, **wearing dark blue and white**, has her right leg straight in the air. |
| | | $h_1$: There is a gymnast. | $(p \rightarrow h_1)$ | |
| | | $h_2$: The gymnast is younger. | $(p \rightarrow h_2)$ | |
| | | $h_3$: The gymnast is wearing dark blue and white. | $(p \not\rightarrow h_3)$ | |
| | | $h_4$: The leg is right. | $(p \rightarrow h_4)$ | |
| | | $h_5$: The leg is straight in the air. | $(p \rightarrow h_5)$ | |
| | | $h_6$: The gymnast has a leg. | $(p \rightarrow h_6)$ | |
| EA |  | $h$: A child wearing blue floaties leaning over at the beach. | $(p \rightarrow h)$ | Words suggested: blue floaties<br>(Original hypothesis with highlight explanation) A child wearing **blue floaties** leaning over at the beach. |
| | | $h_1$: There is a child. | $(p \rightarrow h_1)$ | |
| | | $h_2$: There are some floaties. | $(p \perp h_2)$ | |
| | | $h_3$: There is a beach. | $(p \rightarrow h_3)$ | |
| | | $h_4$: The floaties are blue. | $(p \perp h_4)$ | |
| | | $h_5$: The child is wearing floaties. | $(p \rightarrow h_5)$ | |
| | | $h_6$: The child is leaning over at the beach. | $(p \rightarrow h_6)$ | |
| R | A man sitting on steps with a child as a bystander walks by. | $h$: A man and a child sitting on steps saw a bystander. | $(p \rightarrow h)$ | Words suggested: saw<br>(Original hypothesis with highlight explanation) A man and a child sitting on steps **saw** a bystander. |
| | | $h'_1$: There is a dude. | $(p \rightarrow h'_1)$ | |
| | | $h'_2$: There is a baby. | $(p \rightarrow h'_2)$ | |
| | | $h'_3$: There are staircase. | $(p \rightarrow h'_3)$ | |
| | | $h'_4$: There is a eyewitness. | $(p \rightarrow h'_4)$ | |
| | | $h'_5$: The dude is sitting on steps. | $(p \rightarrow h'_5)$ | |
| | | $h'_6$: The baby is sitting on steps. | $(p \rightarrow h'_6)$ | |
| | | $h'_7$: The dude saw bystander. | $(p \perp h'_7)$ | |
| ER | An elderly man in a tie die shirt playing a red according key bored for a boy in blue. | $h$: There is a man playing an instrument. | $(p \rightarrow h)$ | Words suggested: playing instrument<br>(Original hypothesis with highlight explanation) There is a man **playing** an **instrument**. |
| | | $\neg h_1$: There is not a man. | $(p \not\rightarrow \neg h_1)$ | |
| | | $\neg h_2$: There is not an instrument. | $(p \rightarrow \neg h_2)$ | |
| | | $\neg h_3$: The man is not playing instrument. | $(p \rightarrow \neg h_3)$ | |
| ERE | A female dancer wearing all white dancing while a photographer wearing all white takes her picture. | $h$: A girl is dancing and another man take a photo. | $(p \rightarrow h)$ | Words suggested: man take photo<br>(Original hypothesis with highlight explanation) A girl is dancing and another **man take** a **photo**. |
| | | $h_1$: There is a girl. | $(p \rightarrow h_1)$ | |
| | | $h_2$: There is a man. | $(p \perp h_2)$ | |
| | | $h_3$: There is a photo. | $(p \perp h_3)$ | |
| | | $h_4$: The girl is dancing. | $(p \rightarrow h_4)$ | |
| | | $h_5$: The man take a photo. | $(p \perp h_5)$ | |

**Table 4**
Number of different types of explanations reported by MTEE.

| MRs | E | A | EA | R | ER | ERE | Total |
|---|---|---|---|---|---|---|---|
| $MR_{dec}$ | 60 | 1428 | 41 | 1149 | 473 | 23 | 3174 |
| $MR_{dec\text{-}neg}$ | 89 | 2334 | 275 | 2191 | 1501 | 280 | 6670 |
| $MR_{dec\text{-}syn}$ | 289 | 1407 | 644 | 850 | 1580 | 929 | 5699 |

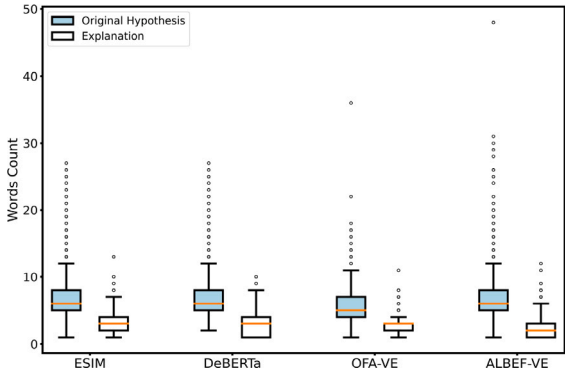



**Fig. 6.** The distribution of lengths of explanations generated by MTEE.

relies on relevant tuples to identify critical information, it may fail to extract the critical information if no relevant tuples can be analyzed.

**Answer to RQ2.** MTEE is able to generate six types of explanations. In the experiments, it successfully generates explanations for 82.9% MR violations, providing useful information to pinpoint the root cause of detected failures. Moreover, these explanations maintain satisfactory compactness.

### 5.3. RQ3 (Performance comparison of subject models)

Based on the evaluation and explanation results, we compare the performance of subject models, and further discuss factors affecting their performance.

We first compare the performance of subject models with respect to the three MRs. We have the following findings.

- Compared with two VE models, two TE models demonstrate better performance. As shown in Table 1, the violation rates on the two VE models are higher than those on the two TE models, for all MRs. This indicates that more failures are revealed from the two VE models.

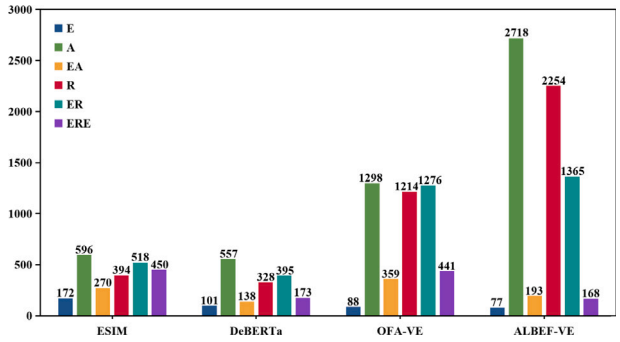



**Fig. 7.** Number of different types of explanations for subject models.

- For the two TE models, DeBERTa performs better than ESIM. For all three MRs, DeBERTa always exhibits lower violation rates than ESIM, with the differences in violation rates ranging from 3.16% to 10.55%.
- For the two VE models, OFA-VE outperforms ALBEF-VE with respect to $MR_{dec}$, while ALBEF-VE performs better with respect to the other two MRs.

Next, we investigate factors affecting the performance of subject models. We collect and summarize different types of explanations for each of the subject models, as depicted in Fig. 7. Based on these, we have the following findings.

- For the four models, the failure in reasoning about attribute-related information is the primary cause of the model's poor performance. In Fig. 7, it can be found that for each subject model, the largest number of explanations relate to type ⟨A⟩, which contains words or phrases describing specific attributes of certain entities.
- Subject models also face the challenge of handling entity and relation information. As can be observed from Fig. 7, a considerable number of explanations describe relation (⟨R⟩) or entity and relation ⟨ER⟩ information.
- Compared to the other types of information, subject models perform relatively better for reasoning about entity information. For each model, only a small number of failures arise from the inference of entity information.

**Answer to RQ3.** Our results suggest that among subject models, TE models generally perform better than VE models, according to the three MRs. It is also revealed that most of the detected failures come from the attribute-related description in the hypothesis, for both TE and VE models.

### 5.4. Threats to validity

In this section, we discuss possible threats to the validity of our study and show how we mitigate them.

The main internal threat to our study refers to the quality of the generated sub-hypothesis. The sub-hypothesis should have consistent facts or information with the original hypothesis, otherwise, it may lead to unreliable evaluation results. To reduce this threat, we have filtered out those incompatible sub-hypotheses during the hypothesis decomposition procedure. In addition, we also employed existing NLP tools to assist the construction of sub-hypotheses. Another internal threat is that MTEE may fail to generate explanations for some violations. Due to the significant variations in textual hypotheses, including differences in complexity and style, extracting critical information from any given hypothesis can be challenging. We plan to take further steps to address this issue in the future, in order to enhance our approach.

The external threat relates to the diversity of our identified MRs. In this study, we have proposed three MRs, which mainly focus on the model's ability to reason complex hypotheses and simple sub-hypotheses. Our MRs are unable to capture a wide range of functionalities, but they target the core aspect of the reasoning task. Moreover, as clarified in Section 3.2, the proposed MRs can be adapted to entailment models that take a textual hypothesis. Another threat to validity lies in the generality of our approach. To mitigate this threat, we have evaluated our approach on representative TE and VE models, and also used publicly available datasets. Given that datasets used in our experiments may not fully capture the complexity of real-life entailment scenarios, we plan to conduct more comprehensive evaluations in the future, in order to increase the practical impact of our approach.

**Table 5**
The length of original hypothesis and explanations.

| Entailment models | | Original hypothesis | | | Explanation | | |
|---|---|---|---|---|---|---|---|
| | | Avg. | Min | Max | Avg. | Min | Max |
| TE | ESIM | 6.78 | 1 | 27 | 3.09 | 1 | 13 |
| | DeBERTa | 6.76 | 2 | 27 | 2.80 | 1 | 10 |
| VE | OFA-VE | 5.65 | 1 | 36 | 2.67 | 1 | 11 |
| | ALBEF-VE | 6.59 | 1 | 48 | 2.51 | 1 | 12 |

## 6. Discussion

In this section, we discuss the characteristics as well as limitations of MTEE.

At first, MTEE complements previous MT-based approaches specifically designed for TE or VE. As reported in [10], the violation rates of MRs for TE models (ESIM and DeBERTa) range from 1.6% to 13.3%, whereas our proposed MRs achieve significantly higher violation rates, spanning 10.36% to 48.57% (see Table 1). For MRs tailored to VE, the overall violation rates are 47.4% for OFA-VE and 55.2% for ALBEF-VE [11]. While our MRs exhibit slightly lower overall violation rates (39.30% and 33.46%, as shown in Table 1), their effectiveness varies across different MRs, with each set excelling in specific aspects. Table 6 further provides two examples highlighting the specificity of our MRs as compared to TE- and VE-specific MRs. Notably, our MRs uncover failures that remain undetected by existing TE and VE MRs, demonstrating their complementary role in enhancing test effectiveness.

Secondly, MTEE can be adapted to a broad range of entailment tasks. Beyond traditional visual entailment and textual entailment, emerging research has extended entailment tasks to audio entailment (i.e., predicting whether an audio premise logically implies a textual hypothesis) and video entailment (i.e., determining if a video sequence supports a given textual hypothesis). Although these entailment tasks target at varying forms of premise, they still employ a textual hypothesis. Since our proposed MRs operate merely on the textual hypothesis of an input, MTEE can be easily adapted to audio and video entailment models.

Thirdly, our proposed MRs can be adapted or extended to other intelligent models. Regarding the ability to support model explanations, the key characteristic of the proposed MRs lies in the whole-part relationship between the original hypothesis and its follow-up hypotheses, which is resulted from the decomposition operations employed by the MR. It should be noted that the decomposition operation can be applied to various data types, such as textual and image data. From this perspective, the input operation of our proposed MRs is adaptable to other models or systems. Nevertheless, different systems implement

**Table 6**
Case studies of applying TE-specific, VE-specific, and our proposed MR.

(a) Applying a TE-specific MR (denoted by $MR_{gender}$) and $MR_{dec}$ by using the same original input, for evaluating ESIM model. $MR_{gender}$ constructs follow-up inputs by changing gender-relevant terms in the original premise and hypothesis, and requires the consistent entailment results between the original and follow-up inputs [10].

| Original input | Follow-up input of $MR_{gender}$ | Follow-up input of $MR_{dec}$ |
|---|---|---|
| $(p, h)$<br>$p$: A woman in a brown shirt is kneeling in the street.<br>$h$: A woman is on the ground outside.<br><br>Original output : $p \rightarrow h$ | $(p', h')$<br>$p'$: A man in a brown shirt is kneeling in the street.<br>$h'$: A man is on the ground outside.<br><br>Follow-up output : $p' \rightarrow h'$<br>(satisfaction) | $(p, h_1)$, $(p, h_2)$, $(p, h_3)$<br>$h_1$: There is a woman.<br>$h_2$: The woman is on the ground.<br>$h_3$: The woman is outside.<br>Follow-up output :<br>$p \rightarrow h_1$, $p \not\rightarrow h_2$, $p \rightarrow h_3$<br>(violation) |

(b) Applying a VE-specific MR (denoted by $MR_{erase}$) and $MR_{dec}$ by using the same original input, for evaluating OFA-VE model. $MR_{erase}$ constructs follow-up inputs by erasing an un-linked object in the original premise, and requires the consistent entailment results between the original and follow-up inputs [11].

| Original input | Follow-up input of $MR_{erase}$ | Follow-up input of $MR_{dec}$ |
|---|---|---|
| $(p, h)$<br>$p$:  <br>$h$: A girl and woman look at their reflection.<br>Original output : $p \rightarrow h$ | $(p', h)$<br>$p'$:  <br><br>Follow-up output : $p' \rightarrow h$<br>(satisfaction) | $(p, h_1)$, $(p, h_2)$, $(p, h_3)$, $(p, h_4)$, $(p, h_5)$<br>$h_1$: There is a girl.<br>$h_2$: There is a woman.<br>$h_3$: There is a reflection.<br>$h_4$: The girl looks at the reflection.<br>$h_5$: The woman looks at the reflection.<br>Follow-up output :<br>$p \rightarrow h_1$, $p \rightarrow h_2$, $p \rightarrow h_3$,<br>$p \rightarrow h_4$, $p \perp h_5$<br>(violation) |

varying functionalities, so the expected output relationships may vary even when the same decomposition-based input operation is applied. In other words, our proposed MRs can be extended to other systems, but may require appropriate adaptation or configuration with respect to the system under test.

Last but not least, MTEE can be a good basis for improving entailment models. On one hand, the discovered failures can serve as a valuable new data source for model retraining or fine-tuning. In this study, several failures were detected, accompanied by a considerable number of premise-hypothesis pairs. By carefully re-labeling these pairs, a new dataset can be constructed. This dataset can either be used to augment the original dataset for retraining a model, or be employed in the process of fine-turning, with the aim of improving or optimizing the performance of the entailment model. On the other hand, MTEE provides local explanations for each identified failures, and thus it enables explanation-based debugging. For example, explanations can guide the model to focus on specific input elements during training [34], and can also be leveraged in a human-in-the-loop setting to help mitigate errors in a trained model [35].

Nevertheless, MTEE still has several limitations. To ensure broader generality, the current implementation relies solely on hypothesis-centric MRs. While empirically effective, this approach may overlook premise-related failures. Future improvements could address this limitation by integrating more diverse MRs into the framework. Furthermore, to generate high-quality sub-hypotheses, we implement MTEE using ChatGPT. However, this dependency may constrain flexibility, especially during service unavailability. To mitigate this limitation, we plan to investigate offline NLP methods that can deliver similar performance in sub-hypothesis generation.

## 7. Related work

*Evaluation methods for TE and VE models.* There have been a lot of evaluation approaches proposed to evaluate the entailment models. Naik et al. [36] proposed the stress test for TE in order to explore the extent to which these models understand the semantic content of sentences. Stress test assessed the robustness of the TE models for specific linguistic phenomena such as word overlap, antonym, length mismatch, etc. Wang et al. [37] proposed TextFlint, a robustness evaluation toolkit for NLP tasks including TE, which incorporates universal text transformation, adversarial attack, and subpopulation. Stowe et al. [38] created a dataset consisting of paired sentences to evaluate the ability of TE models to understand figurative language. Salvatore et al. [39] use equivalent transformations such as synonym replacement to evaluate TE models. Different from the above approaches, our approach, MTEE, evaluates both TE and VE models, by following the pipeline of metamorphic testing.

*Metamorphic testing of TE and VE models.* Our previous work [10] applied MT to evaluate TE models, which solved the oracle problem and focused on functionalities such as monotonicity reasoning, lexical inference, conjunctive sentences, etc. It effectively revealed failures for different TE models and the MT results are able to interpret TE models' capabilities from varying aspects. As a follow-up study, this paper makes several new contributions. On one hand, three innovative MRs are proposed for both TE and VE, serving as the basis for conducting MT. Different from previous MRs that consider some linguistic phenomenon and logic inference, our MRs mainly focus on the entailment of a complete hypothesis and a set of its sub-hypotheses. On the other hand, our proposed MT-based approach facilitates not only quality evaluation, but also failure explanation. Recently, Li et al. [40] employed MT to discern bias in TE tasks performed by Large Language Models (LLMs). By transforming textual inputs with nuanced demographic attributes such as sex, race, occupation, age, and socioeconomic status, MT has revealed the potential biases of the LLMs. Chang et al. [11] proposed VEglue, which adopts MT to test the VE models. The proposed MRs are based on the alignment information between the image premise and textual hypothesis. Our approach, MTEE, is also driven by MT. Nevertheless, we proposed MRs that are generic to both TE and VE, and our approach supports both failure detection and explanation.

*Explanation approaches for TE and VE models.* A considerable amount of effort has been put into the explanation approaches for TE and VE models. Camburu et al. [41] proposed e-SNLI, a dataset with natural language explanations for the ground truth labels extended from the SNLI. The explanations were crowd-sourced using Amazon Mechanical Turk. Kumar et al. [42] proposed a TE model that generates faithful nature language explanations, which were generated by using a pre-trained language model. Chen et al. [43] incorporated counterfactual

examples, original examples, and external knowledge as the input for TE model. They employed a knowledge-aware generative pre-trained language model to generate contrastive explanations, which help clarify the differences between varying predictions. For VE, Do et al. [44] introduced the e-SNLI-VE corpus, which was formed by appending human-written NLEs to SNLI-VE. They also extended a VE model with the capacity to learn from these explanations at training time and then provide an explanation for each predicted label. Yang et al. [45] used an attention-based reasoning network in VE models to integrate token-level and chunk-level vision–language representations. Then they incorporated the focused words or chunks into explanation generation. In this study, we propose a post-hoc explanation method to identify the information that is critical for the detected failures. Our approach does not rely on the internal information of the model, and it can be applied to both TE and VE models.

## 8. Conclusion and future work

In this paper, we proposed a unified metamorphic testing framework for evaluating and explaining textual entailment (TE) and visual entailment (VE) models. With the observations in the commonality and characteristics of TE and VE, we identified three MRs that are generic to both tasks. Specifically, our MRs perform text decomposition on the original hypothesis of an input to construct follow-up hypotheses, while they keep the premise unchanged in both original and follow-up inputs. Based on this, our approach conducts metamorphic testing to detect failures of TE and VE models. Additionally, once a failure is detected, it further conducts a post-hoc analysis within the relevant group of inputs, to identify parts of information in the original hypothesis that is critical to the detected failure. The experimental results showed that the proposed approach is effective for detecting and explaining failures for TE and VE models, and is also helpful for revealing the performance characteristics of subject models. In the future, we plan to extend the proposed MRs to cover a broader range of functionalities of entailment tasks, so as to further improve the failure detection effectiveness. Furthermore, we will also enhance the post-hoc analysis, in order to improve the effectiveness of model explanation.

## CRediT authorship contribution statement

**Mingyue Jiang:** Project administration, Methodology, Data curation, Conceptualization. **Bintao Hu:** Software, Data curation. **Xiao-Yi Zhang:** Writing – review & editing, Methodology, Conceptualization.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT and DeepSeek in order to improve the readability. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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